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Project Proposal: FleetFlow

Odoo x Gujarat Vidhyapith Hackathon

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1. Team Information

Team Name	404BrainRotNotFound
Frontend Developer	Divy Choksi
Backend & DB Admin	Dhruv Pandya
Tester & Validator	Naivedi Binjwa
Documentation	Jay Gajjar

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2. Problem Statement

Chosen Problem

FleetFlow: Modular Fleet & Logistics Management System — A centralized digital hub designed to replace inefficient manual logbooks and optimize the lifecycle of delivery fleets.

Problem Analysis & Current Challenges

- **Inefficient Manual Coordination:** Reliance on paper logbooks leads to data entry errors and lost records.
- **Poor Visibility:** Dispatchers lack real-time data on vehicle availability, leading to idle assets.
- **Safety Compliance Gaps:** Manual tracking often misses expired licenses, creating legal liabilities and safety risks.
- **Reactive Maintenance:** A lack of data leads to "run-to-failure" maintenance instead of preventative care, increasing long-term costs.
- **Financial Blind Spots:** Inability to accurately calculate Cost-Per-KM or ROI per vehicle.

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3. Proposed Solution Statement

FleetFlow is a centralized, rule-based digital hub designed to replace inefficient, manual logbooks with an intelligent fleet management ecosystem. The solution provides a "Command Center" that optimizes the entire lifecycle of a delivery fleet by integrating real-time asset tracking, automated safety compliance, and financial performance monitoring into a single interface.

Key Capabilities

- **Smart Dispatching:** Automatically validates cargo weight against vehicle capacity to prevent overloading.
- **Automated Maintenance Logic:** Instantly locks vehicles out of the dispatch pool when service is logged, preventing unsafe usage.
- **Financial Intelligence:** Calculates precise ROI and operational costs per vehicle to drive data-backed decisions.

Uniqueness

- **Technical Innovation:** Integration of Odoo’s ORM for strict relational constraints between Expenses, Trips, and Vehicle IDs.
- **Operational Benefits:** Ensures 100% compliance by dispatching zero trips with expired licenses.
- **ROI Clarity:** Provides exact calculation of profitability per asset.

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4. Motivation

The primary motivation is to eliminate the risks and inefficiencies associated with manual fleet management.

- **Safety & Compliance Risks:** Without automated checks, drivers with expired licenses or vehicles due for maintenance are often assigned to active trips.
- **Operational Inefficiency:** Manual dispatching results in overloaded vehicles or idle assets due to lack of visibility.
- **Financial Blind Spots:** Owners struggle to calculate true "Cost-Per-Kilometer" or ROI because fuel and maintenance data are not linked to specific trips.

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5. Target Audience

FleetFlow is designed for four specific user roles with tailored access permissions:

- **Fleet Managers:** Responsible for vehicle health, asset lifecycle, and scheduling maintenance.
- **Dispatchers:** Operational users who create trips, assign drivers, and validate cargo loads.
- **Safety Officers:** Compliance monitors who track license expirations and analyze "Safety Scores".
- **Financial Analysts:** Business users who audit fuel spending, calculate maintenance ROI, and review cost reports.

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6. Solution Overview

FleetFlow is an Odoo-based modular platform that revolutionizes logistics through intelligent rule-based automation.

System Workflow

The system operates on a cyclic workflow ensuring safety compliance and data integrity.

1. **Vehicle Intake & Registration:** A manager inputs a new vehicle (e.g., "Van-05") and defines attributes like max load capacity. The system initializes status as "Available".
2. **Driver Compliance Check:** When a driver is selected, the system runs an automatic background check. It verifies the license validity for the selected vehicle category; if expired, the assignment is blocked.
3. **Smart Dispatching & Validation:** The dispatcher assigns a driver and cargo. The system compares cargo weight against max capacity ($Cargo < MaxCapacity$). If passed, Vehicle and Driver status update to "On Trip".
4. **Trip Completion:** Upon reaching the destination, the driver marks the trip as "Done" and enters the final odometer reading. The system releases assets back to "Available".
5. **Maintenance Lockout Protocol:** If a service event is logged, the vehicle status automatically switches to "In Shop" and is hidden from the dispatch pool .

6. **Analytics & Financial Updates:** Fuel and maintenance costs are recorded. The system recalculates "Cost-per-km" and updates ROI metrics using the formula

$$ROI = \frac{Revenue - (Maintenance + Fuel)}{AcquisitionCost}$$

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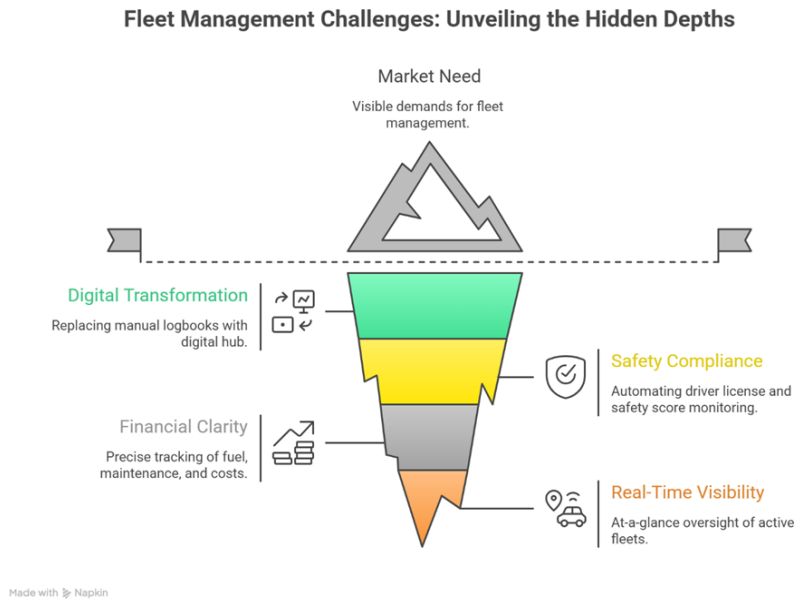
7. Market Need & Requirements

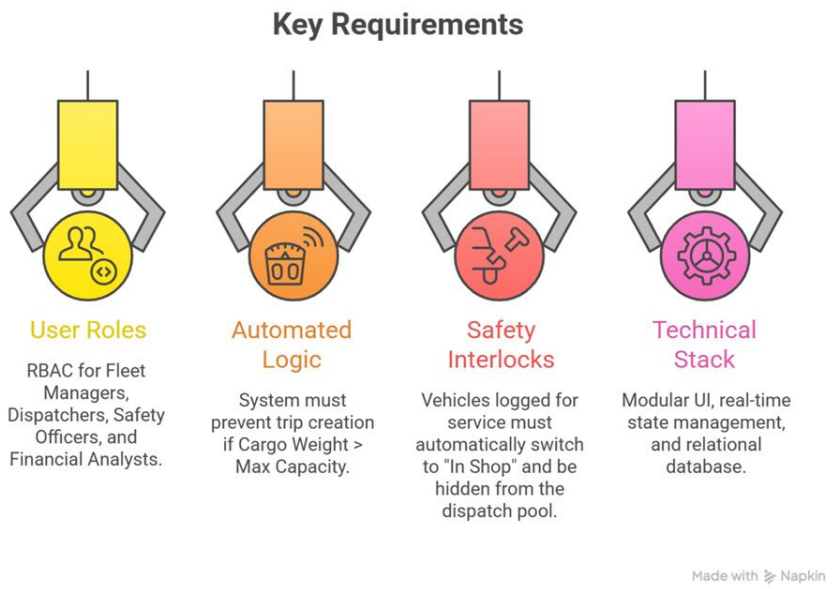
Market Need

- **Digital Transformation:** Urgent need to replace manual logbooks with a centralized digital hub.
- **Safety Compliance:** Demand for automated systems to monitor licenses and safety scores to prevent liability.
- **Financial Clarity:** Requirement for precise tracking of fuel spend and operational costs per vehicle.
- **Real-Time Visibility:** Need for "at-a-glance" oversight of active fleets to prevent idle assets.



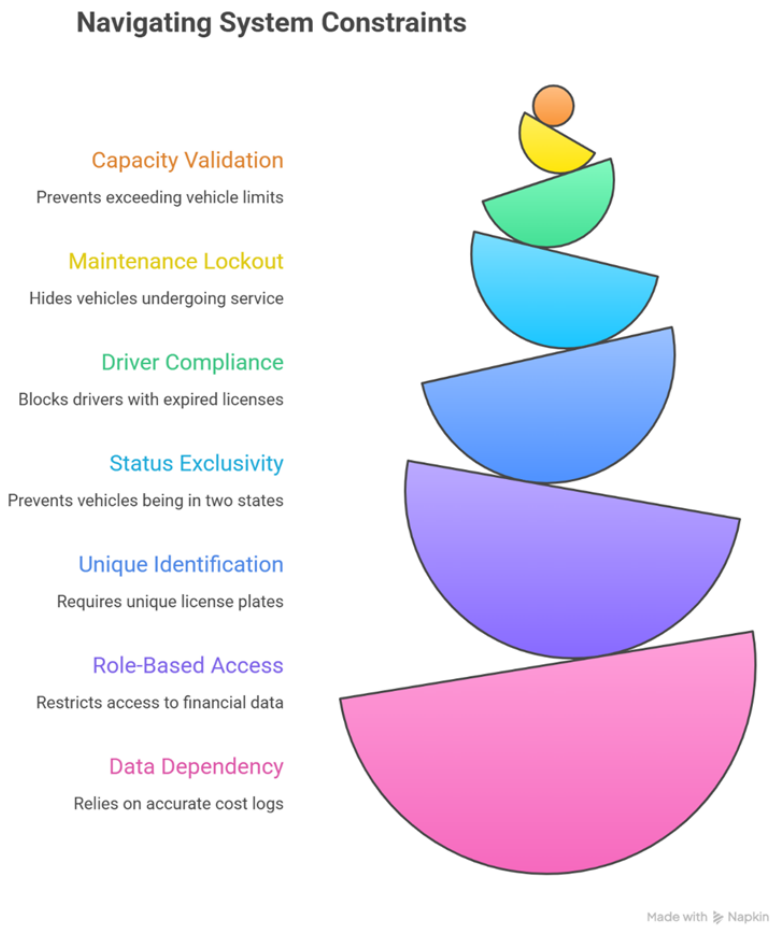
Key Requirements

- **User Roles:** Strict Role-Based Access Control (RBAC).
- **Automated Logic:** System must prevent trip creation if Cargo Weight > Max Capacity.
- **Safety Interlocks:** Vehicles logged for service must automatically switch to "In Shop".
- **Technical Stack:** Modular UI with scannable data tables, real-time state management, and a relational database .



8. Assumptions & Constraints

- **Capacity Validation:** Prevents exceeding vehicle limits.
- **Maintenance Lockout:** Hides vehicles undergoing service.
- **Driver Compliance:** Blocks drivers with expired licenses.
- **Status Exclusivity:** Prevents vehicles from being in two states simultaneously.
- **Unique Identification:** Requires unique license plates.
- **Role-Based Access:** Restricts access to financial data.



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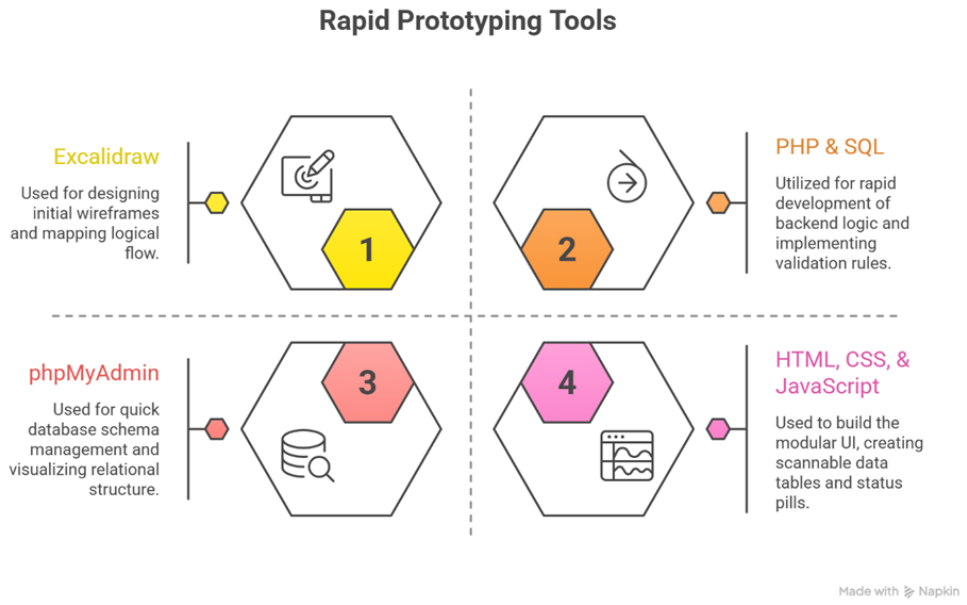
9. Tools Utilised for Rapid Prototyping

- **Excalidraw:** Used for designing initial wireframes and mapping logical flow.
- **PHP & SQL:** Utilized for rapid development of backend logic and implementing validation rules.
- **phpMyAdmin:** Used for quick database schema management and visualizing relational structure.

- **HTML, CSS, & JavaScript:** Used to build the modular UI, creating scannable data tables and status pills.



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10. Feasibility & Implementation

1. Technical Feasibility

- **Proven Stack:** PHP and SQL ensure a stable backend capable of handling complex relational data.
- **Rapid Development:** HTML, CSS, and JS allow for quick creation of a modular, lightweight UI.
- **Data Integrity:** Relational structure enforces unique constraints (e.g., License Plates) to prevent duplicate records.

2. Operational Feasibility

- **Safety & Compliance:** Automates safety checks to reduce legal liability.
- **Efficiency Gains:** Reduces dispatching time and optimizes vehicle utilization rates.
- **Maintenance Logic:** Shifts operations from reactive repairs to proactive asset management via automated lockouts.

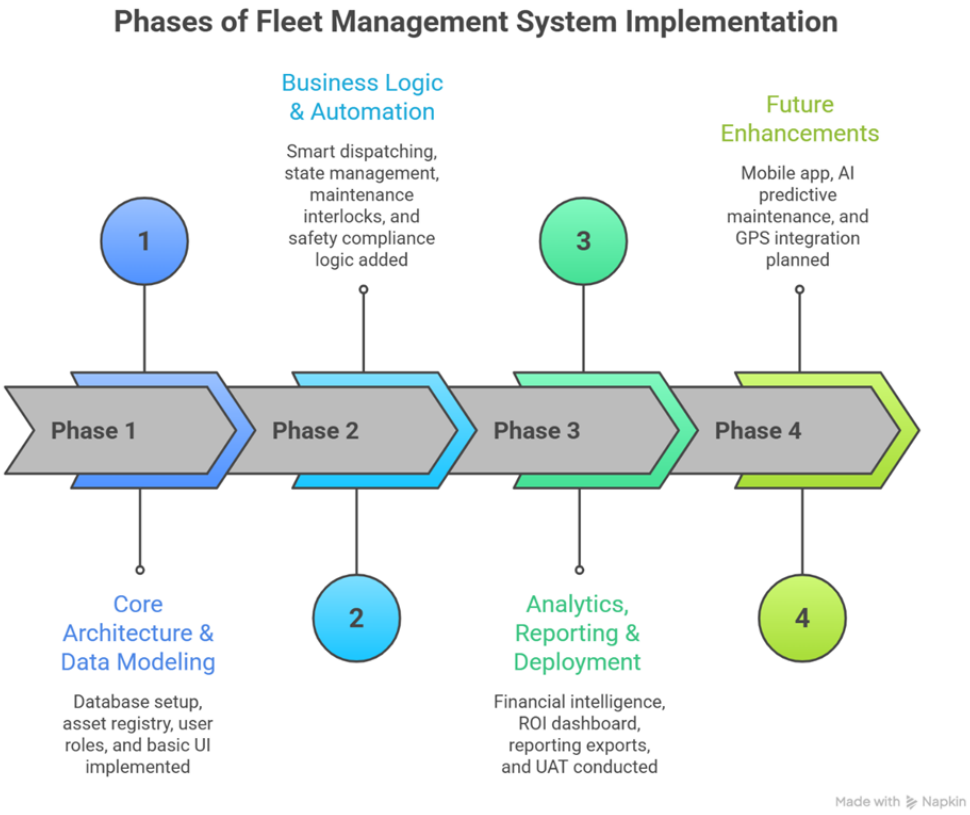
3. Economic Feasibility

- **Cost Reduction:** Preventative maintenance tracking helps avoid costly major breakdowns.

- **ROI Clarity:** Precise "Cost-Per-KM" calculation enables data-driven decisions to maximize profitability.

Phases of Implementation

1. **Core Architecture & Asset Registry:** Establishing database structure, RBAC, and CRUD operations.
2. **Business Logic & Automation:** Implementing "Smart Dispatching" validation and automated maintenance interlocks.
3. **Analytics & Reporting:** Developing the financial intelligence engine for ROI calculations and audit exports.
4. **Future Enhancements:** Mobile app for driver inputs, AI predictive maintenance, and GPS integration.



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11. Frameworks/Technologies

Technology Stack

Layer	Technology Used
Frontend Layer	Odoo QWeb
Backend Layer	Python (Odoo ORM)
Infrastructure	Odoo.sh

Selection Reasoning

- **Odoo ORM:** Efficiently handles complex relational data (Vehicles $\leftrightarrow$ Trips $\leftrightarrow$ Expenses).
- **PostgreSQL:** Ensures ACID compliance for financial transaction integrity.
- **Modular Architecture:** Allows easy addition of features without breaking core logic.
- **QWeb Views:** Enables rapid UI development using native view inheritance.

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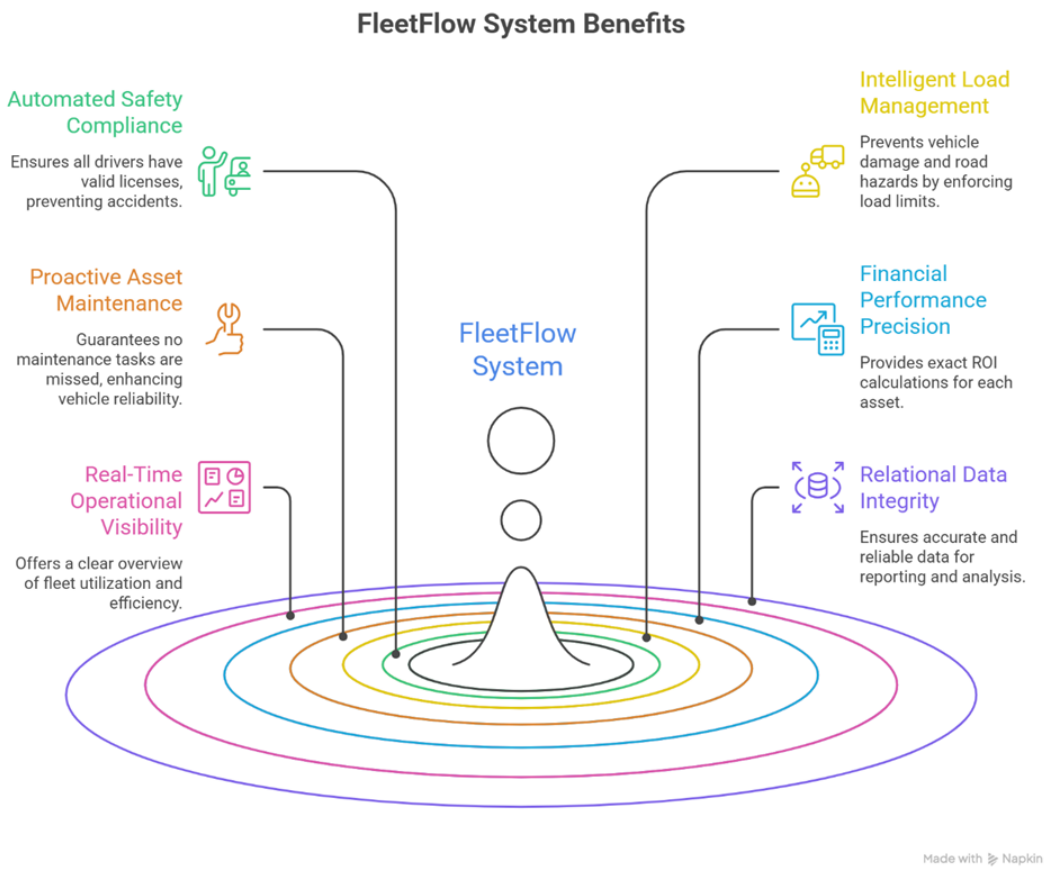
12. Effectiveness

- **Automated Safety Compliance:** Ensures all drivers have valid licenses, preventing accidents.
- **Proactive Asset Maintenance:** Guarantees no maintenance tasks are missed.
- **Real-Time Operational Visibility:** Offers a clear overview of fleet utilization.
- **Intelligent Load Management:** Prevents vehicle damage by enforcing load limits.
- **Financial Performance Precision:** Provides exact ROI calculations for each asset.



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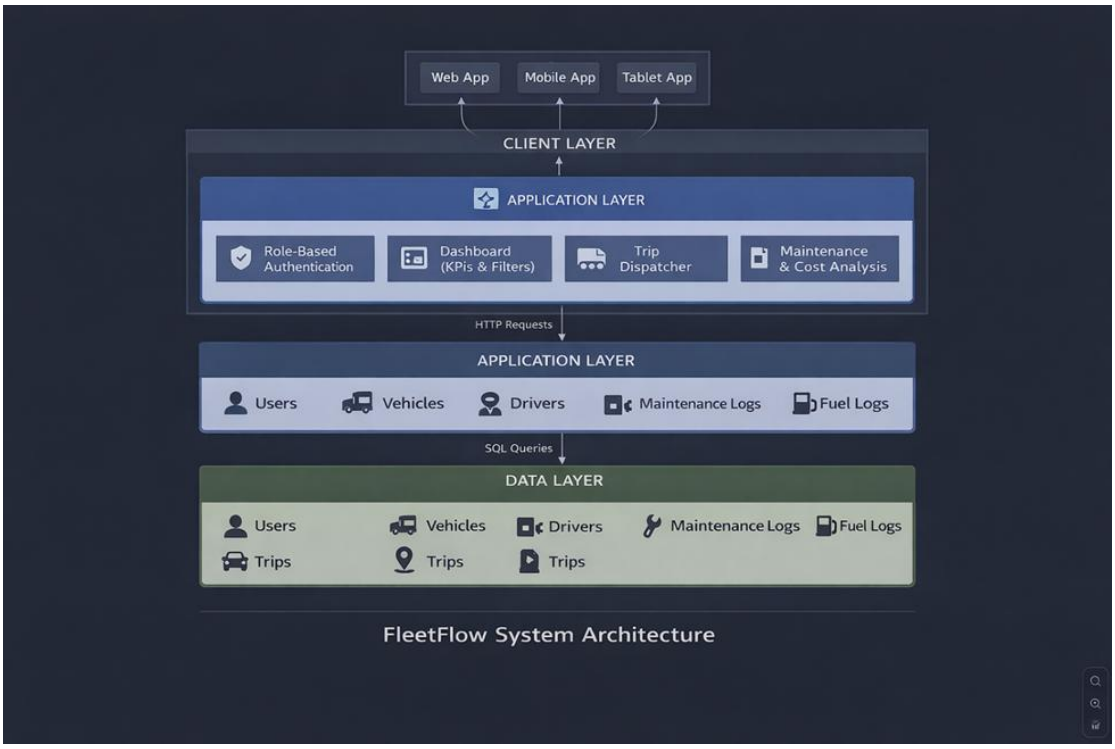
13. UI/UX Mockup - Screens Overview

1. **Authentication Flow:** Role-based login and secure session management.
2. **Command Center (Dashboard):** KPI cards for Active Fleet Count, Utilization, and Pending Cargo.
3. **Dispatch Workflow:** Input sequence (Select Vehicle $\rightarrow$ Driver $\rightarrow$ Cargo) with visual feedback (Green "Success" pill / Red "Error" pill).
4. **Analytics & Reporting:** Visual graphs showing Fuel Efficiency (km/L) and Cost Analysis with PDF export buttons.



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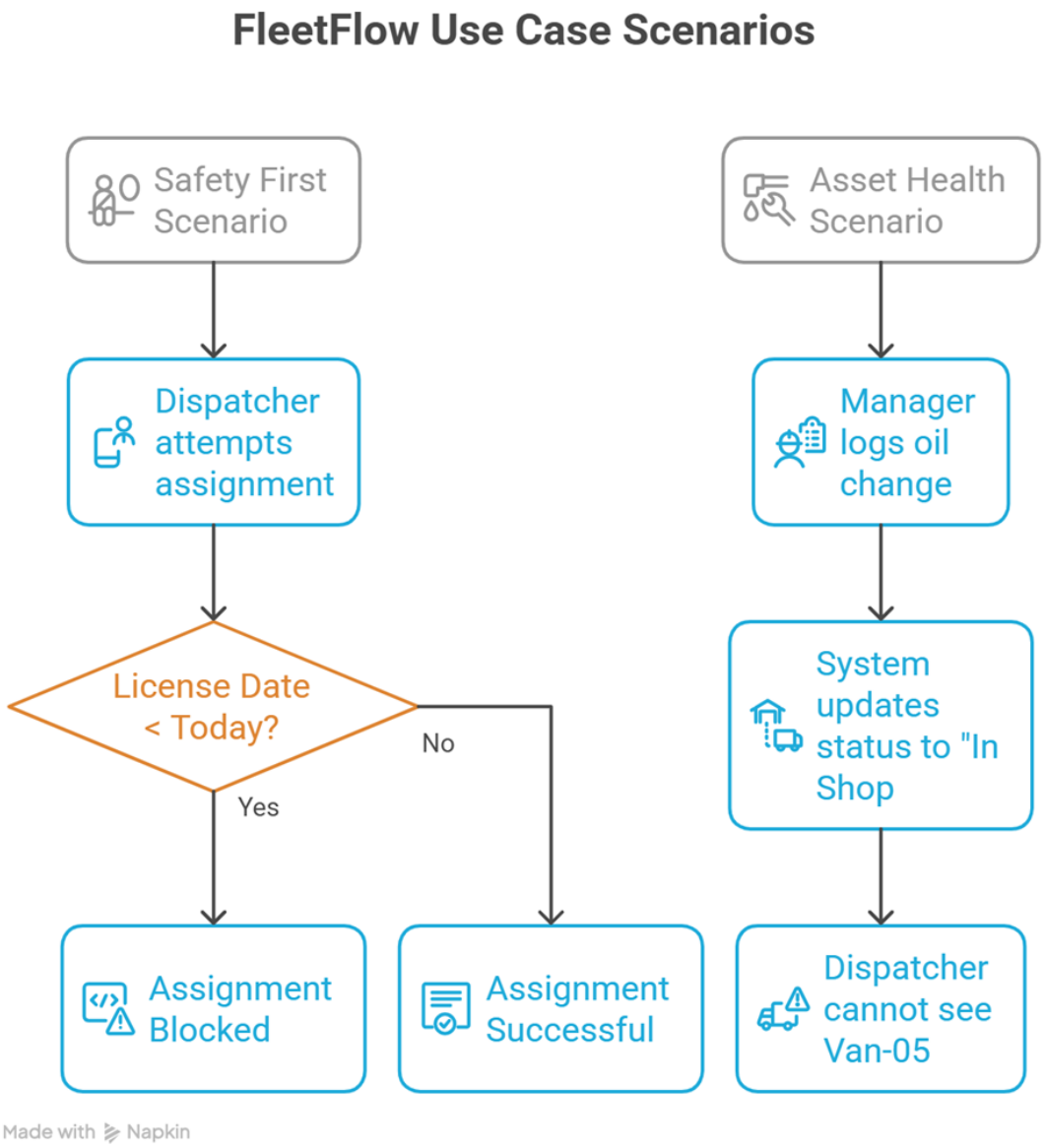
14. Business Scope and Use Case

Scenario 1: "Safety First"

- **Situation:** A driver ("Alex") has an expired license.
- **Action:** Dispatcher attempts to assign Alex to a delivery.
- **System Logic:** Detects License Date < Today.
- **Result:** Assignment is blocked; legal compliance is maintained.

Scenario 2: "Asset Health"

- **Situation:** Manager logs an "Oil Change" for Van-05.
- **Action:** System immediately updates status to "In Shop".
- **Result:** Dispatcher cannot see Van-05 in the list, preventing unsafe usage.



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15. System Design and Architecture

Backend Services (Odoo Module Structure)

- **Models:** fleet_vehicle.py, fleet_trip.py, fleet_driver.py, fleet_expense.py.
- **Views:** fleet_dashboard.xml, fleet_vehicle_view.xml, fleet_trip_view.xml.
- **Controllers:** main.py.
- **Security:** ir.model.access.csv.

Coding Approach

- **Agile Implementation:** Iterative development allowing concurrent testing of frontend and backend logic.
- **Sprint Structure:** One accelerated hackathon week including planning, daily standups, and final review .
- **Code Quality:** Strict adherence to database integrity and real-time state management.

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16. Conclusion

FleetFlow represents a critical step forward in digital logistics management. By transitioning from manual, error-prone logbooks to an automated, rule-based ecosystem, the system directly addresses the industry's most pressing challenges: safety compliance, operational efficiency, and financial transparency. With its robust Odoo-based architecture, "Smart Dispatching" logic, and real-time analytics, FleetFlow not only prevents costly liabilities but also empowers fleet owners with the data-driven insights needed to maximize the ROI of every asset on the road. The solution is technically feasible, economically viable, and ready for rapid deployment to transform fleet operations.